

CBSE Class 10 Mathematics - Answer Key

1. 1. Find the sum of the first 15 terms of the AP 20, 24, 28, ...

Answer: The required sum is 720.

Solution:

1. Here $a = 20$, $d = 4$ and $n = 15$.
2. Use $\frac{n}{2}[2a + (n - 1)d]$ to calculate the sum.
3. $S_{15} = \frac{15}{2}[40 + 14 \times 4] = 720$.

Derivation:

$$S_n = \frac{n}{2}[2a + (n - 1)d]$$
$$S_{15} = \frac{15}{2}[2(20) + (15 - 1)4]$$
$$S_{15} = 720$$

Arithmetic Progressions — sum of n terms of an AP

2. 2. Solve the quadratic equation $x^2 - 5x + 6 = 0$ and verify the roots.

Answer: The roots are $x = 2$ and $x = 3$.

Solution:

1. Factorise the polynomial: $x^2 - 5x + 6 = (x - 2)(x - 3)$.
2. Set each factor equal to zero.
3. Therefore $x = 2$ or $x = 3$.

Derivation:

$$x^2 - 5x + 6 = 0$$
$$(x - 2)(x - 3) = 0 = 2, 3$$

Quadratic Equations — factorisation method